Return Pilot

From scattered return information
to one clear next action.

AIPI 590 · Mid-Semester Presentation

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Purchase receipt 02

Sample retailer B

Everyday Hoodie

Order **RP-1028**

Purchased **Sep 20, 2026**

Delivery date
Not provided

Total **$49.99**

When is it due?

Multiple purchases. Multiple retailers. Different rules.

I know what I want to return.

But what do I actually do next?

Order email 01

Sample retailer A

Wireless Headphones

Order **RP-1042**

Delivered **Sep 15, 2026**

Order confirmed **$59.99**

Purchase screenshot 03

Sample retailer A

Phone Case

Order **RP-1042** · **$19.01**

Delivered **Sep 15, 2026**

Still deciding

Where do I return it?

Retailer A · Return policy

Start within **30 days of delivery**

Check the method.
Prepare the matching label.

A / Headphones + phone case

What should I do first?

Retailer B · Return policy

Window starts on **delivery**

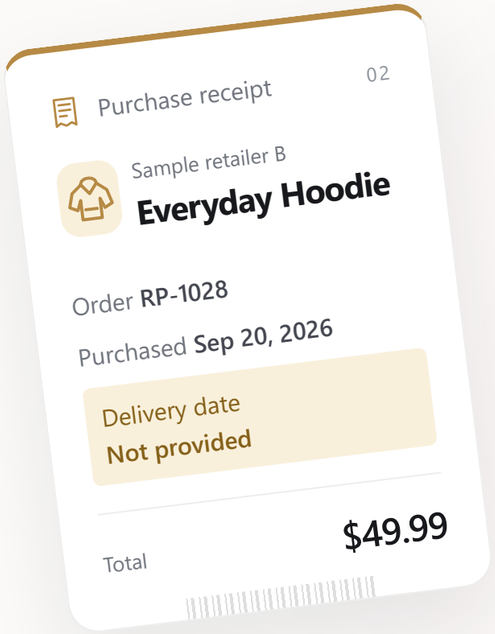
Proof of purchase required.
Confirm the date first.

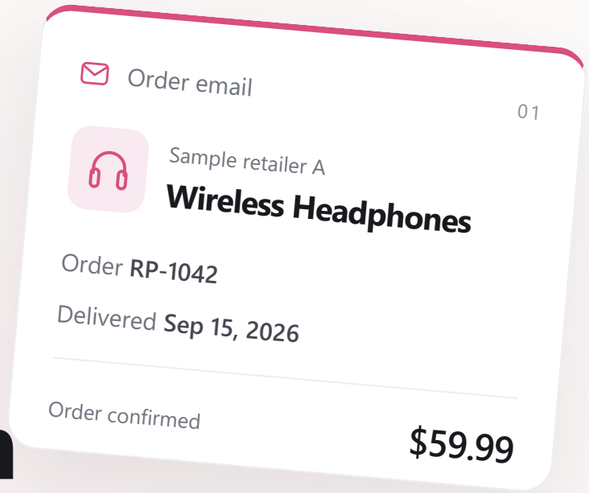
B / Everyday Hoodie

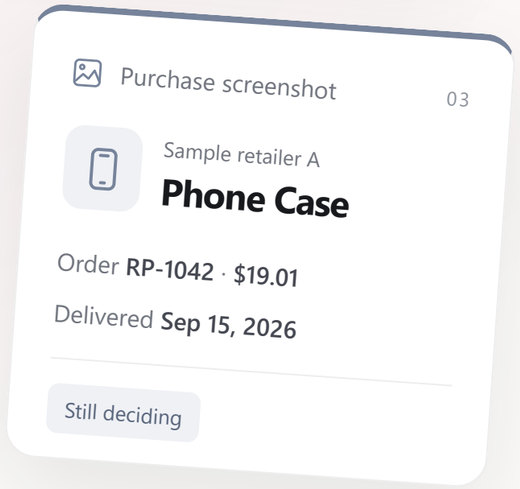
Illustrative example

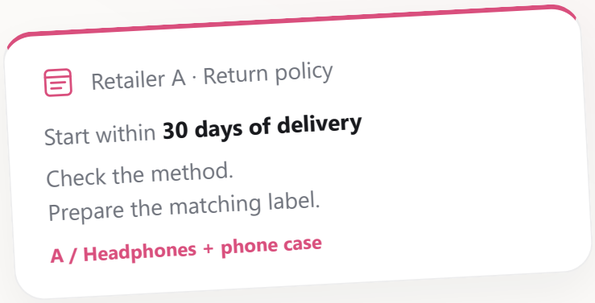
Same purchases. Same policies. Less manual work.

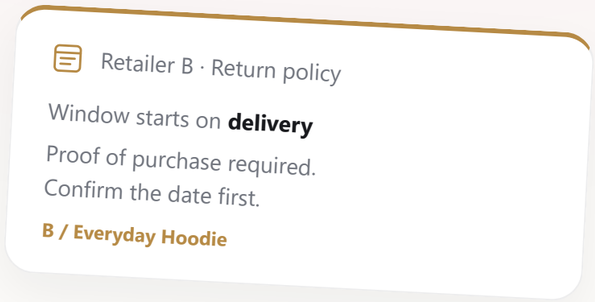
We turn return information into next actions.









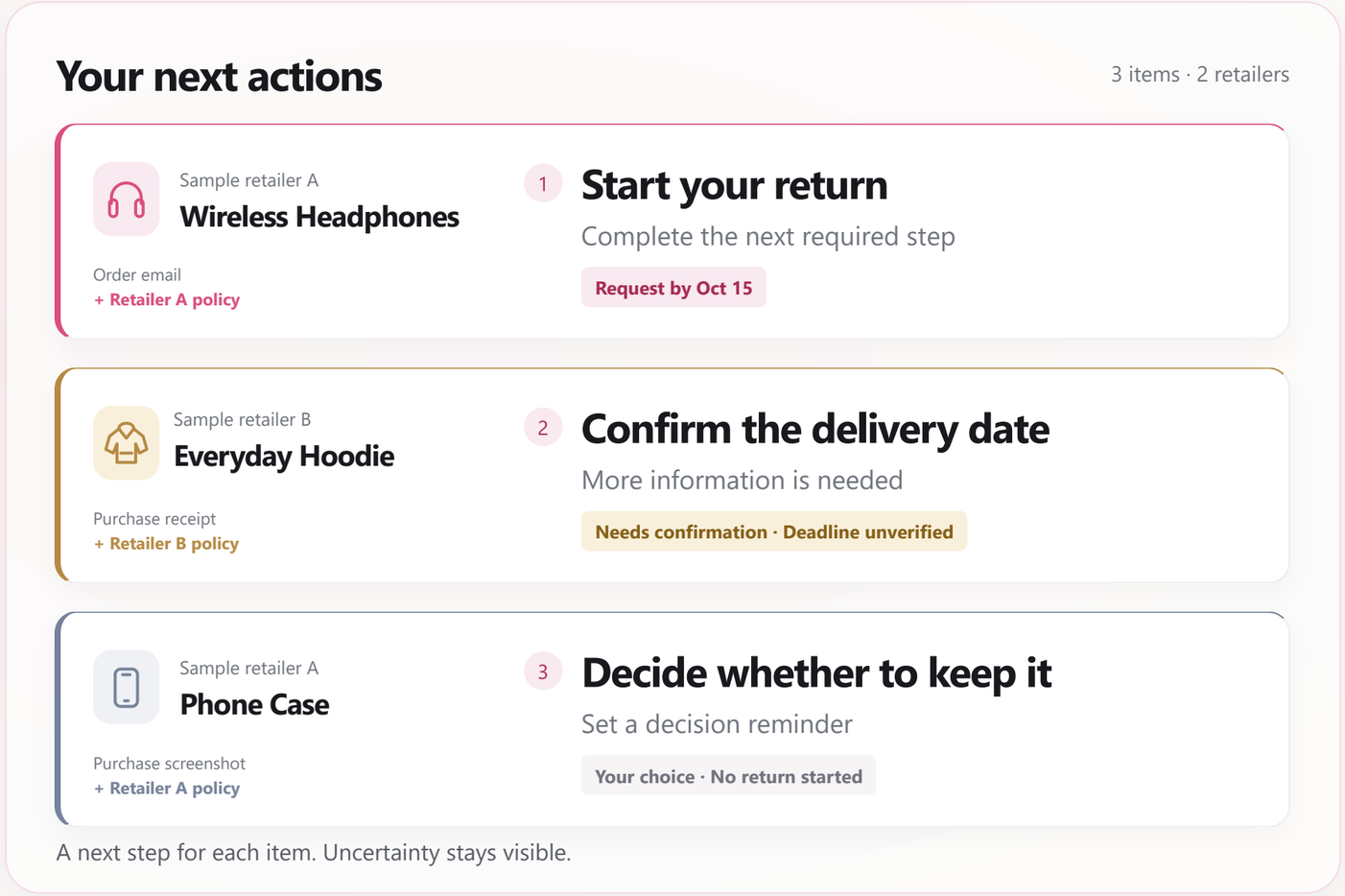


Illustrative example

Return Pilot

One place to manage returns across retailers

Less searching.
Less remembering.
A clear next action.



A next step for each item. Uncertainty stays visible.

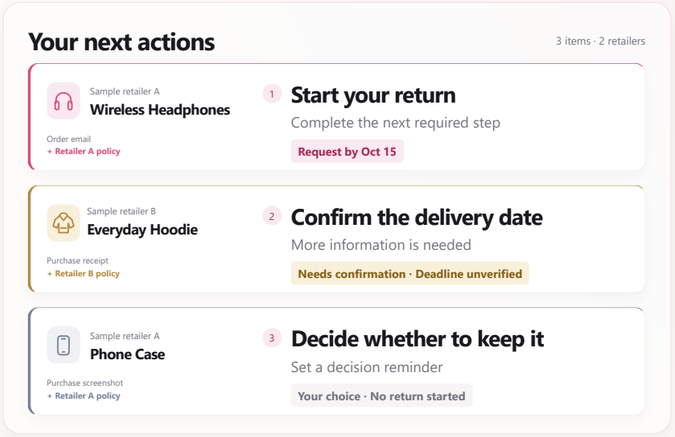
Illustrative example

Progress since our idea pitch

THE ORIGINAL IDEA

One place to manage returns across retailers

Experience concept



Illustrative example

01 Product scope

Scope to confirm

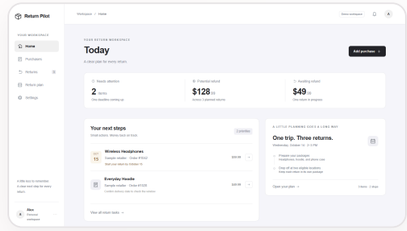
Prototype flow mapped · Retailer coverage / conditions to confirm

02 Product experience

Purchase information → Tasks → Action plan

Web prototype

Open Web demo ↗



Actual prototype capture · Sample data

03 AI evaluation

20 cases tested

Which approach fits which task?

Rules | LLM | Hybrid

✓ Planning-layer experiment complete

20 fixed cases · Detailed results next

Less searching. A clearer next step.

The pain: each return starts with another search through receipts, emails, and retailer rules.

E

Emma

Graduate student

Shops across several retailers.

"I'm really happy with this. I finally know what to return first, without searching all my emails and store policies."

Expected response · Satisfied

The same purchases. A different experience.

Information → Action

Your next actions

1 **Wireless Headphones**

Start your return

Request by October 15 · Based on delivery date

2 **Everyday Hoodie**

Confirm the delivery date

Deadline stays unverified until the date is confirmed

3 **Phone Case**

Decide whether to keep it

The choice stays with you · No return started

Design value: **one structured task list**, with priorities and uncertainty visible.

Illustrative persona · Expected feedback to validate

Useful—if I control what I share.

The concern: managing a few returns should not require access to an entire inbox.

R

Ryan

Software engineer

Careful about sharing personal data.

"I'd use it, but I don't want an app reading my entire inbox just to manage a few returns."

Expected response · Cautious

Privacy concern → A narrower input boundary

Proposed architecture

My inbox

- Personal messages
- Work conversations
- Order confirmation
- Travel plans
- Subscriptions

✓ No full-inbox permission required.

Only the purchases you choose

- Forward one order email
- Upload a receipt
- Enter details manually

Selected evidence → Purchase understanding → Return tasks

Architecture response: **selective purchase input**. Wider security controls still need implementation.

Illustrative persona · Input boundary is a planned design, not a security certification

Plan on the web. Follow through on the go.

The request: make the next step accessible where returns actually happen.

M

Maya

Working professional

Fits errands around work and commuting.

"I'd love this on my phone—especially when I'm at the drop-off point and need the instructions."

Expected request · Mobile access

From planning at home to action outside

Future opportunity

Return Pilot · Web workspace

One trip. Three returns.

Organize the tasks before leaving home.

Wireless Headphones Request return

Everyday Hoodie Confirm date

Current focus · Web MVP

→

Same tasks and progress

←

Mobile concept

At the drop-off?

Keep the next step within reach.

Open return instructions

Photograph the receipt

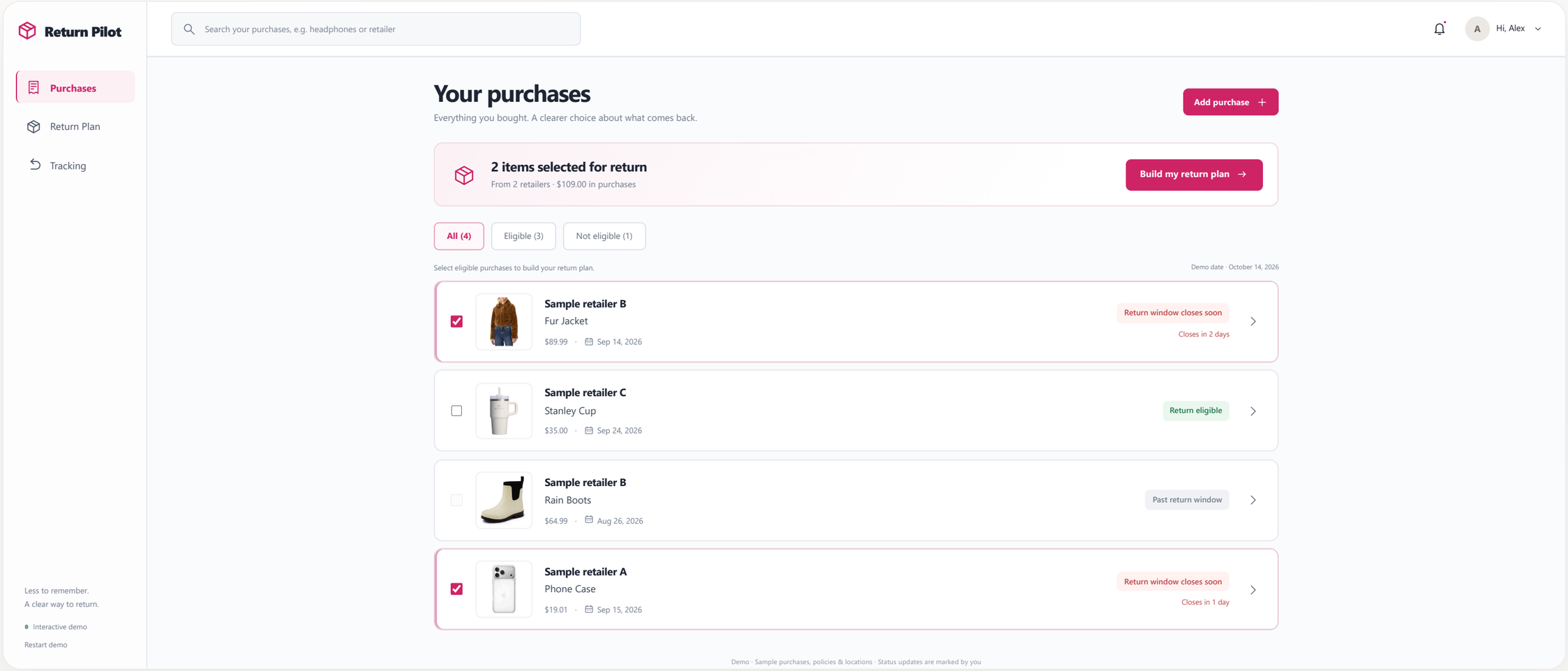
Update progress

Value · Clear next actions | **Trust** · Selective input | **Next** · Mobile access

Illustrative persona · Mobile is a future extension, not a delivered feature

Our web MVP

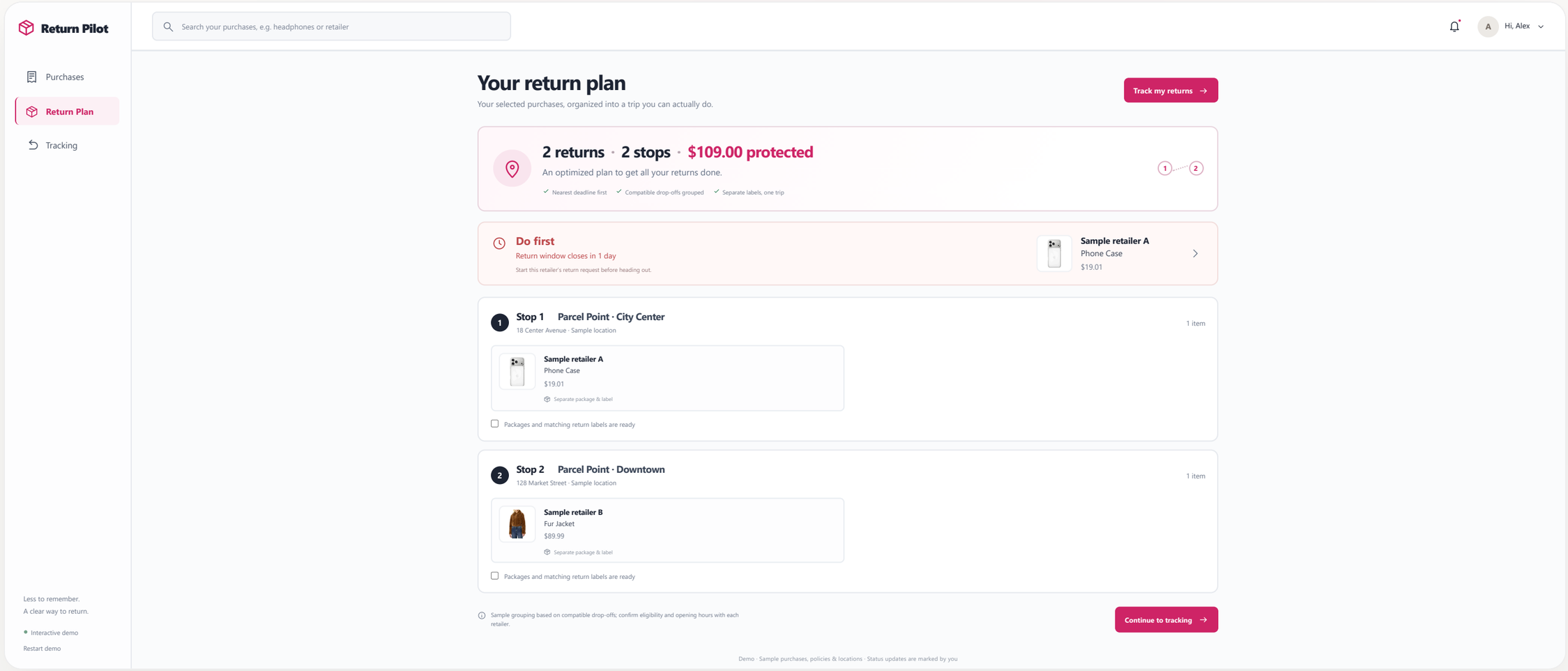
Collect purchases. Make one plan. Follow every refund.



Interactive demo · Sample receipt extraction, policies and drop-off locations · Manual tracking

Our web MVP

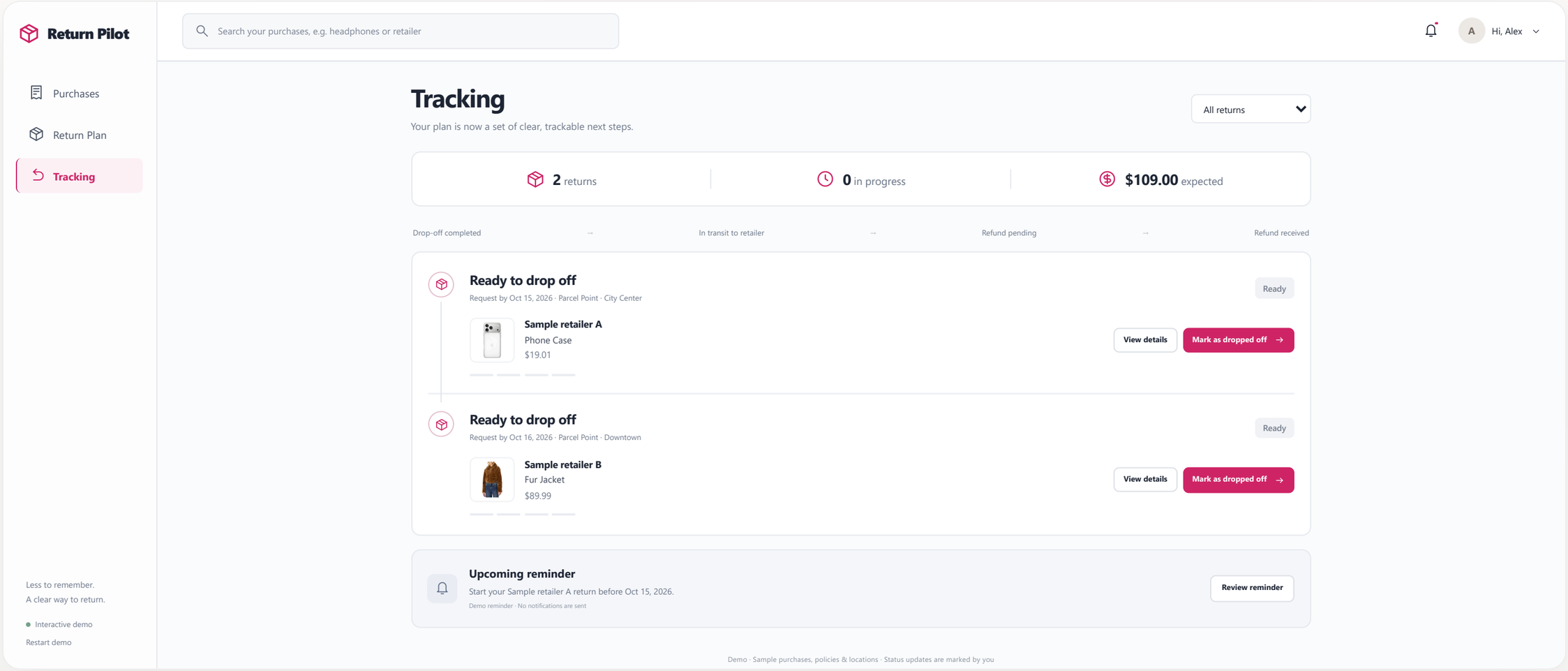
Collect purchases. Make one plan. Follow every refund.



Interactive demo · Sample receipt extraction, policies and drop-off locations · Manual tracking

Our web MVP

Collect purchases. Make one plan. Follow every refund.



Interactive demo · Sample receipt extraction, policies and drop-off locations · Manual tracking

From purchase to refund.

AI interprets. Rules calculate. You decide.

01

Import

Receipt / PDF
Email / Manual

AI Extract details

02

Match policy

Check conditions
Calculate deadlines

AI Assess eligibility

03

Choose

You select items
Urgent returns first

Your decision

04

Plan

Time + place
Group nearby returns

AI Use preferences

05

Return

Steps, labels & QR
You drop it off

Guided by the app

06

Track refund

Email updates
You confirm refund

AI Read status

POLICY LIBRARY

AI checks updates → Team reviews → **Policies go live**

Updates checked weekly

Cloud MVP → On-device AI next

Planned: encrypted transfer & storage

Monthly budget for 200 active users.

Cloud MVP · Estimated **$40–70** per month

≈7 model calls per complete return

2	1	3	1
Import & policy	Plan	Email updates	Adjust / retry

Monthly calls per user

2 × purchases + 5 × returns

Usage per user / month	Platform / month
Light 10 purchases · 2 returns	**$40**
Typical 20 purchases · 4 returns	**$50**
Heavy 40 purchases · 8 returns	**$70**

Inside the $50

$25	$7	$8	$10
Server, data & email	AI	Search	Spare budget

Estimated service costs · Actual bills will depend on usage

Pricing: GPT-6 Luna ↗ · Brave Search ↗

Experiment: Where does AI actually add value?

Same 20 return-planning cases · Same inputs · Four strategies · Two models

20 **Fixed return-planning cases**

Same purchase context and policy evidence

RULES BASELINE

A

RULES ONLY

Deterministic baseline

LLM-BASED APPROACHES

B0

BASIC PROMPT

LLM baseline

→

B1

STRUCTURED

Decision flow + worked example

→

C

HYBRID

B1 + validation & rule fallback

MODELS

- Qwen 3.7 Flash
- GPT-6 Luna

EVALUATION METRICS

- Correct plan
- Critical errors
- Latency
- Tokens
- Cost

Experiment: Structured prompting improved planning accuracy.

Same models. Same 20 cases. More explicit decision structure.

B0 — Foundational Prompt

What the model should do

Role & Task Framing

Define the assistant's role and planning objective.

Context & Constraints

Set evidence, policy boundaries, and restrictions.

Structured Output

Require a consistent JSON schema.

→

B1 — Structured Prompt

How the model should decide

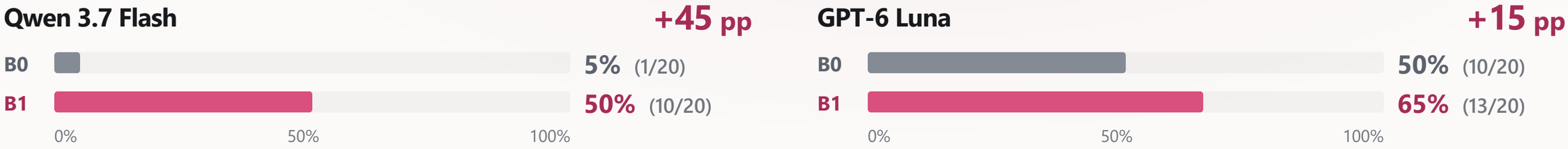
B0 foundation retained · three added techniques

01 **Structured Reasoning**
Explicit internal decision procedure (CoT-style).

02 **Example-Based Prompting**
One complete worked example (one-shot).

03 **Self-Verification**
Prompt-level consistency check before final output.

CORRECT PLANS · B0 VS B1



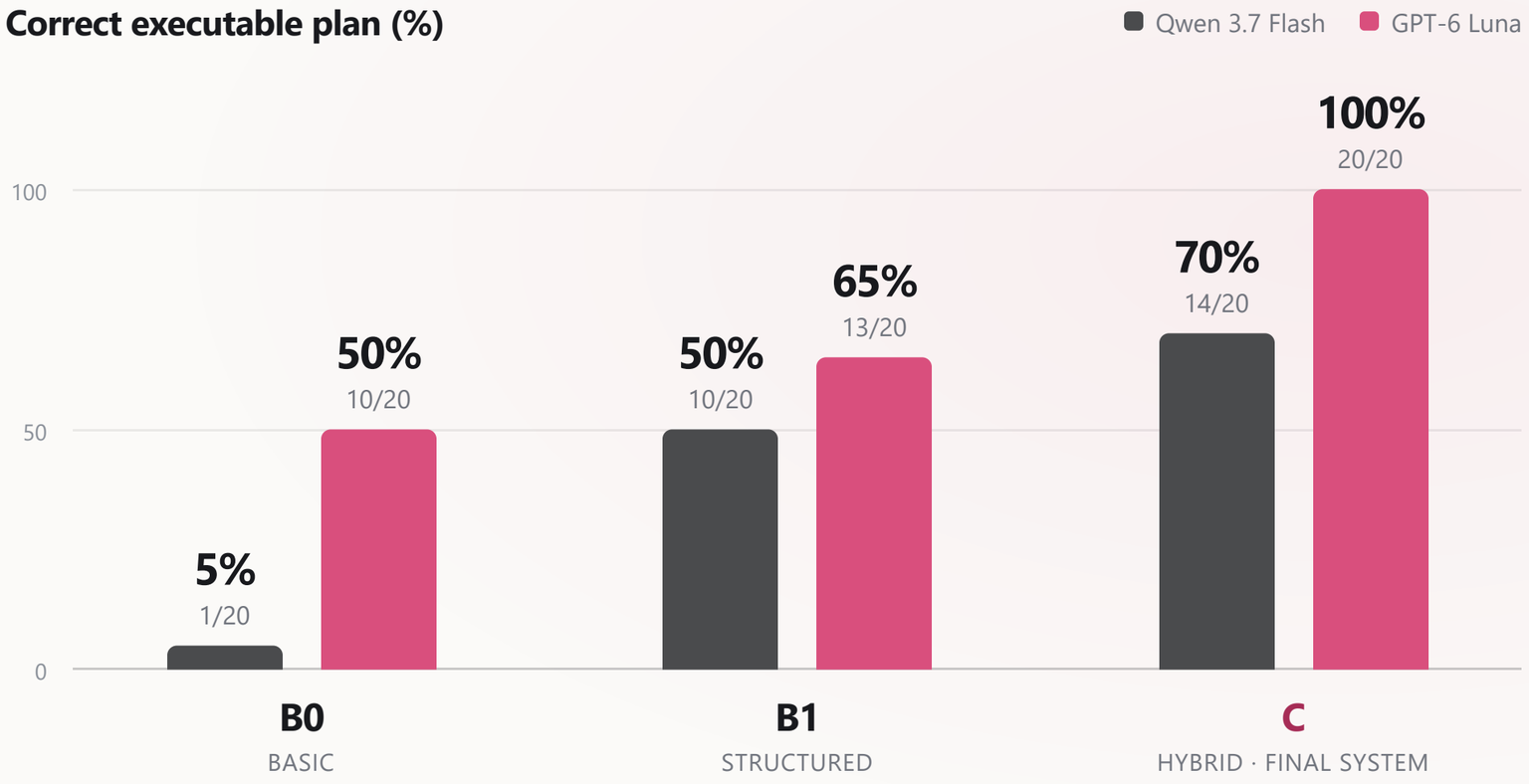
Same models. Same cases. Better decision structure.

pp = percentage points

Experiment: The result changed how we divide the work.

Same 20 cases. Better prompts, then deterministic validation.

Correct executable plan (%)



RULES BASELINE

100% **20 / 20**

No model call

On this bounded deterministic test set

C fallback triggered

Qwen	**8 / 20** · 40%
GPT	**4 / 20** · 20%

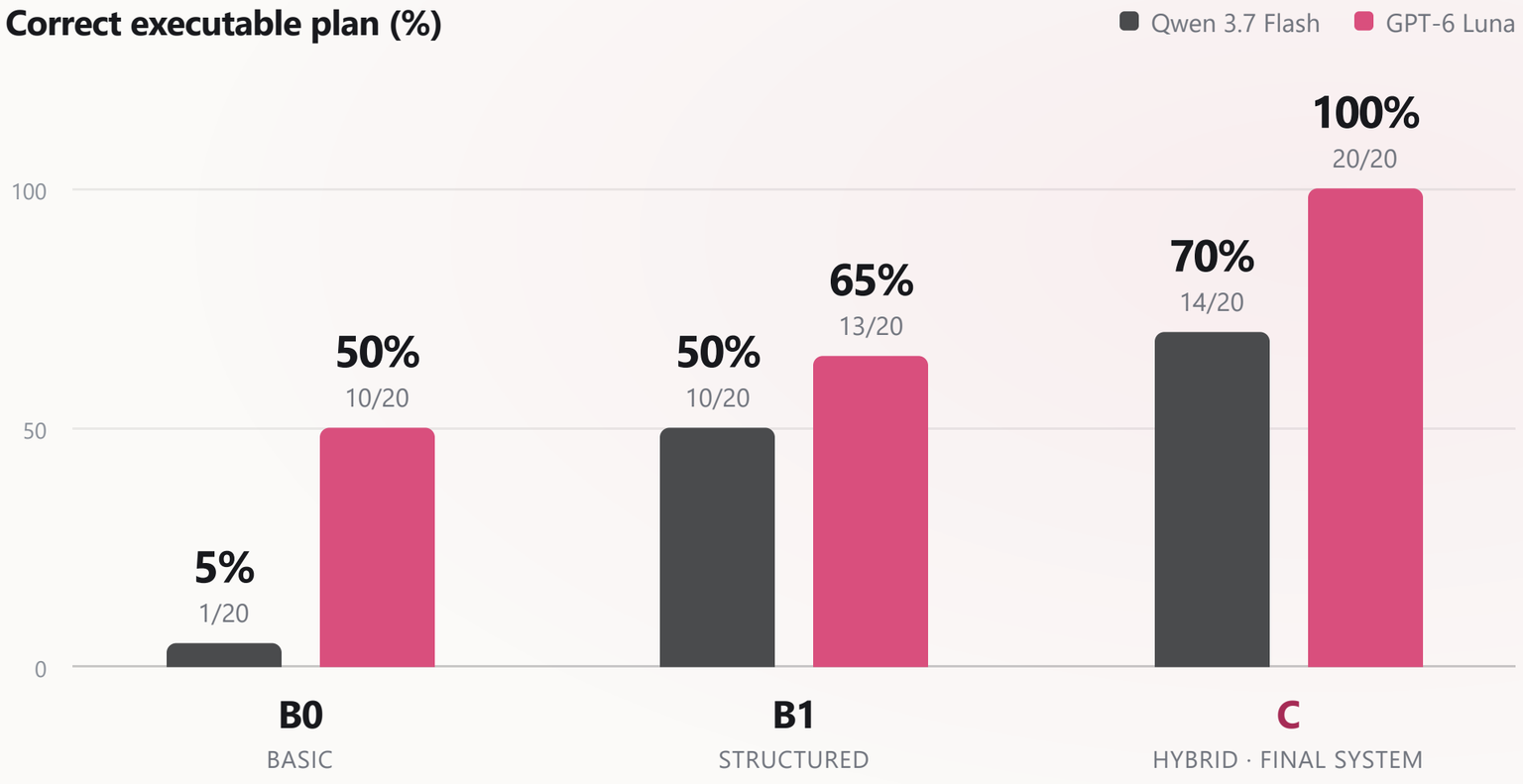
LLM PLAN → **DETERMINISTIC CHECK** → **ACCEPT / RULE FALLBACK**

No second LLM repair call

Experiment: The result changed how we divide the work.

Same 20 cases. Better prompts, then deterministic validation.

Correct executable plan (%)



RULES BASELINE

100% **20 / 20**

No model call

On this bounded deterministic test set

WHAT IT TOOK

34 decision checks

21 scenario-routing branches

7 fallback / exception paths

~116 rule-logic LOC

Nike QR already initiated
→ explicit 21-day handoff rule

Rules are strongest where correctness is computable.

PROMPT Structure improves model behavior → **HYBRID** Validation improves reliability → **RULES** Enforce what is computable

Experiment: Token Economics

Model quality comes with a measurable cost trade-off.

Qwen 3.7 Flash

B1

~16.5K tokens

$0.000570

/ case

→

C

~16.7K tokens

$0.000575

/ case

+$0.000005 / case +202 total tokens

GPT-6 Luna

B1

~16.8K tokens

$0.00189

/ case

→

C

~17.0K tokens

$0.00191

/ case

+$0.00002 / case

GPT ≈ 3.3× Qwen model cost / case

B1: $0.00189 / $0.000570 ≈ 3.32× · C: $0.00191 / $0.000575 ≈ 3.32×

Higher model quality comes at ~3.3× the API cost.

Challenges & Next Priorities

What we learned while building — and where we go next.

CHALLENGES

01 Our initial prompts listed requirements without enough structure, which led to poor model outputs.

02 GPT performed better in our evaluation, but costs about 3.3× more than Qwen and still requires substantial policy context.

03 We had to decide both whom to recruit and which research method would give us the most useful product feedback.

NEXT PRIORITIES

Smarter Agent Harness

Better retrieval, context filtering, and tool use

Send only relevant policy evidence to the model.

Privacy

Narrower access, stronger protection, clearer user controls

Define data boundaries, storage, permissions, and user choice.

Mobile Experience

Bring return instructions and progress tracking on the go

Receipt capture · Quick instructions · Progress updates

Q & A

AIPI Midterm Presentation

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